

HABIBUL RAHMAN QALBI

AI ENGINEER • LLM & AGENTIC AI • RAG SYSTEMS

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PROFESSIONAL SUMMARY

AI Engineer with 4+ years building production software, about 3 of them in production AI: LLM-powered agentic and RAG services in Python (FastAPI, Django), plus React / Next.js frontends. Cut AI service cost by up to **70%** and made it up to **4x faster** than a baseline tool-calling agent. Delivered AI services for enterprise customers including Advantech, working directly with C-level leaders to turn ambiguous requirements into reliable products. Open to remote roles worldwide, hybrid / on-site in Indonesia, or relocation.

TECHNICAL SKILLS

LLM & GenAI: OpenAI, Anthropic (Claude), Google GenAI, VoyageAI, prompt engineering, function calling, structured outputs, evaluation, AI safety (PII scrubbing, prompt-injection prevention, moderation)

Agentic AI & RAG: Agentic AI workflows, multi-agent orchestration, Retrieval-Augmented Generation (RAG), LangChain, LlamaIndex, semantic search, reranking

Vector & Data: Pinecone, ChromaDB, embedding models, MongoDB, PostgreSQL, Redis

ML / Deep Learning: TensorFlow, CNNs, computer vision, image classification & segmentation

Backend & APIs: Python, FastAPI, Django, Django REST Framework, Celery, REST APIs

Frontend: React, Next.js, TypeScript, Redux Toolkit, Tailwind CSS, Material UI, Three.js

MLOps & DevOps: Docker, CI/CD, Git, Azure, RabbitMQ, Apache Airflow

Languages: English, Indonesian

PROFESSIONAL EXPERIENCE

AI Engineer

Nov 2023 - Present

Airconcur (Growise) • Taipei, Taiwan (Remote)

Joined as a freelance AI Module Developer; hired full-time in May 2024.

- Architect and ship agentic and RAG-based AI microservices in Python (FastAPI) with multiple LLMs and embedding models, working directly with C-level leaders.
- Up to **70% lower cost** and **4x faster** than a baseline tool-calling agent, through semantic, prompt and tool-output caching, data canonicalization, and context & memory management.
- Re-engineered the retrieval layer (Voyage AI embeddings, reranking, data cleanup and de-duplication): **~30% lower latency** and **~60% higher reasoning accuracy** vs. earlier versions.
- Built containerized, multi-LLM RAG modules with isolated per-user indexes to prevent cross-user data leakage.
- Added AI safety layers: PII scrubbing, prompt-injection prevention and moderation middleware.
- Delivered AI services for enterprise customers, including Advantech; **2x faster builds** after reworking CI/CD.

Full Stack Developer / Software Engineer

May 2023 - May 2024

PT Mandiri International Technology • South Jakarta, Indonesia

- Owned the full project lifecycle, from requirement gathering to deployment.
- Built customer-facing web apps (React, Next.js) and a cross-platform mobile app on a Django REST backend.
- IoT device-backend communication over HTTP and MQTT: **50% higher pipeline performance** vs. a standard MQTT pipeline (parallel input-stream management) and **70% less device bandwidth** (multi-device connection routing).
- Added authenticated data reporting for site monitoring; tuned ORM and queries to handle **100+ concurrent requests**.

Full Stack & Backend Developer (Healthcare AI)

Aug 2022 - Mar 2023

Institut Teknologi Sepuluh Nopember (ITS) • Freelance • Surabaya, Indonesia

- Built the React-Django app and backend serving the research team's U-Net (TensorFlow) model for ultrasound deep-vein segmentation, with 3D visualization in Three.js (research prototype).
- Developed a model handler and preloader to reduce real-time inference latency.
- Replaced legacy auth with JWT and refactored API viewsets on the iBrain2U platform.

Full Stack Web Developer

Apr 2022 - Apr 2023

LnData Inc. • Surabaya, Indonesia

- React / Django products, secure APIs and Apache Airflow data pipelines for social-media data platforms, with multinational teams.
- Managed deployment servers, monitored system load and resolved security issues.

EDUCATION

B.Eng. in Computer Engineering

2017 - 2021

Institut Teknologi Sepuluh Nopember (ITS) • Surabaya, Indonesia • GPA 3.25 / 4.00

- **Focus:** Artificial Intelligence, Microprocessor & Microcontroller Systems
- **Thesis:** CNN-Based Handwashing Motion Classification, published in Jurnal Teknik ITS 10(2), 2021